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Renal and Genitourinary Considerations in Pediatrics I

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Session Objectives:

1. Identify and describe the patient presentation and most common causes of pyelonephritis in pediatric patients.
2. Describe the clinical symptoms, treatment protocol, and physiologic implications for cryptorchidism in a pediatric patient.
3. Describe the clinical presentation and common causes of Potter Syndrome in a pediatric patient.
4. Understand the clinical presentation of various pediatric GU conditions such as Hypospadias, Epispadias, Chordee, and Hair Tourniquet Syndrome.
5. Understand and describe the clinical difference between phimosis and paraphimosis.
6. Discuss some GU issues in females such as labial adhesions, vaginal tags, and imperforate hymen

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Genitourinary Considerations in Pediatric Patients

- Always consider pediatric genitourinary problems throughout prenatal care if oligohydramnios, renal anomalies, or hydronephrosis are detected on prenatal ultrasounds
- Consider genitourinary problems during the newborn exam if you detect pre-auricular pits (ear tags), palpable abdominal masses (suggestive of hydronephrosis), genitalia anomalies, or defects in the urinary stream
    - If other congenital anomalies are present, have a cautious eye for a GU anomaly as there can be many comorbidities

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Pyelonephritis in Pediatrics

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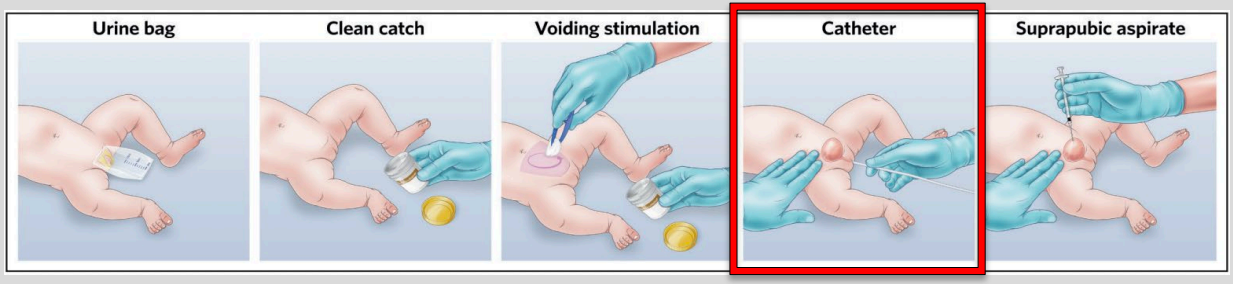
Pyelonephritis: Patient Presentation

- An infant with pyelonephritis may present with the following symptoms:
  - Fever (> 100.4˚F)
  - Irritability
  - Vomiting
  - Diarrhea
  - Lethargy
- Always suspect pyelonephritis if the baby has **systemic symptoms** and a **positive urine culture** (most commonly e. coli)

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Pyelonephritis: Diagnosis in Pediatrics

- Pyelonephritis requires a urine culture to be diagnosed (in both pediatrics and adults)
  - Most often a **catheterized specimen is the diagnostic test of choice**-–it is a sterile technique, accurate, and minimally invasive
  - A suprapubic puncture may be the diagnostic test of choice in premature newborns, but this is invasive and frightening to most parents
  - A clean catch can be suitable for diagnosing pyelonephritis in older children
- Note a **"bag specimen" is NOT appropriate for diagnosis** as the sample can become easily contaminated with the skin flora



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Pyelonephritis: Causes and Treatment

Cause

- Concerns for structural abnormalities (e.g., urinary reflux, posterior urethral valve, and uteropelvic junction obstructions)

Treatment

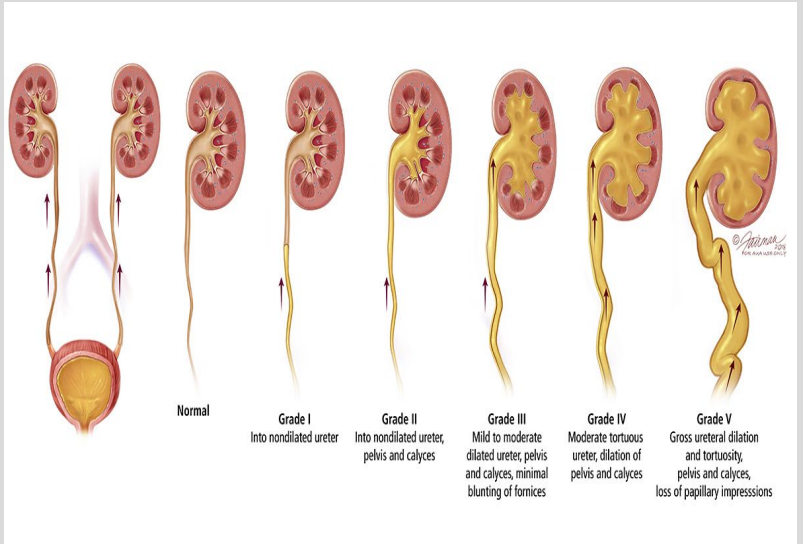
- IV antibiotics specific to pathogen detected (most commonly e. coli)
  - Ceftriaxone
  - Cefotaxime
  - Ampicillin with Gentamicin
- Infants with pyelonephritis clinically improve very quickly if treated immediately

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Pyelonephritis: Vesicoureteral Reflux

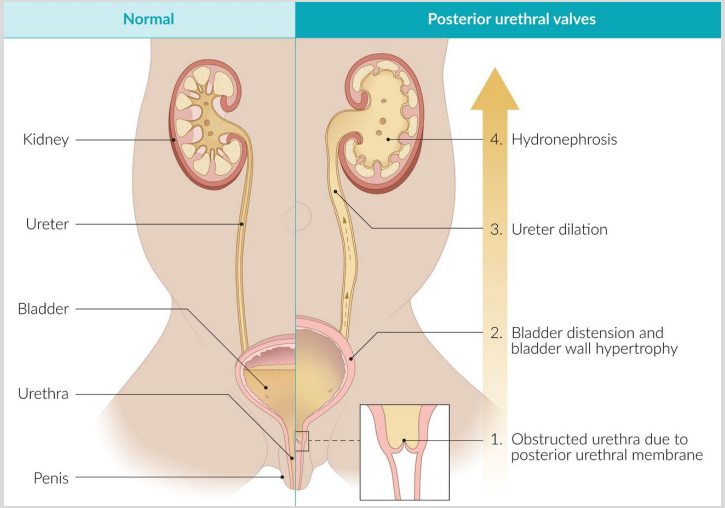
- Vesicoureteral Reflux is characterized as **failure of the bladder to empty properly**
  - Often due to blockage, failure of the bladder muscle, or damage to the nerve that controls normal bladder emptying
  - Buildup of urine in the ureter and kidney can quickly cause infection
- **Grade 1 and Grade 2 urinary reflux is treated with "watch and wait"** vs. Grades 3, 4, and 5 urinary reflux is treated with possible surgical repair and prophylactic antibiotics



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Pyelonephritis: Posterior Urethral Valve

- Posterior urethral valves occur when a **piece of tissue is blocking the urethra**-–often this is not a complete blockage therefore it is associated with a **weak stream**
  - Occurs in **males ONLY**
  - Associated with **bilateral hydronephrosis** and recurrent pyelonephritis due to the back flow of urine
- Can be diagnosed with renal ultrasound +/or VCUG (definitive)
  - Renal ultrasound will show distension of the bladder, ureters, and kidneys

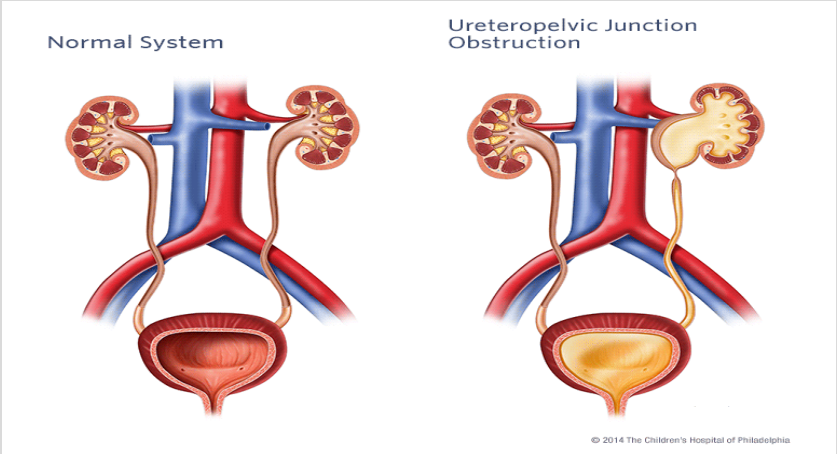


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Pyelonephritis: Uteropelvic Junction Obstruction

- Uteropelvic junction obstruction is associated with **unilateral hydronephrosis** of the ureter and renal pelvis—a kink or anomaly in the junction can cause backflow of urine in the kidney
- This buildup of urine will increase the risk of infection of both the urinary tract and the kidney



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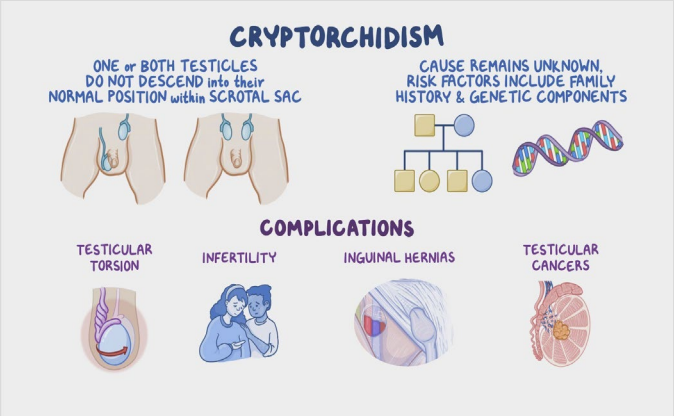
Cryptorchidism

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Cryptorchidism: Overview

- Cryptorchidism occurs when **one of the testicles does not descend within the inguinal canal properly**
  - Affects 1% of the male population but is more common in premature neonates or patients with endocrine/genetic syndromes
- The testicle is not designed to sit close to the abdominal wall (creates a state of overheating)—associated with increased risk of **malignant transformation** (testicular cancer) and **sub/infertility**
- Cryptorchidism is also associated with increased risk of **testicular torsion** and **inguinal hernias**
- All four **complications can occur DESPITE surgical correction**

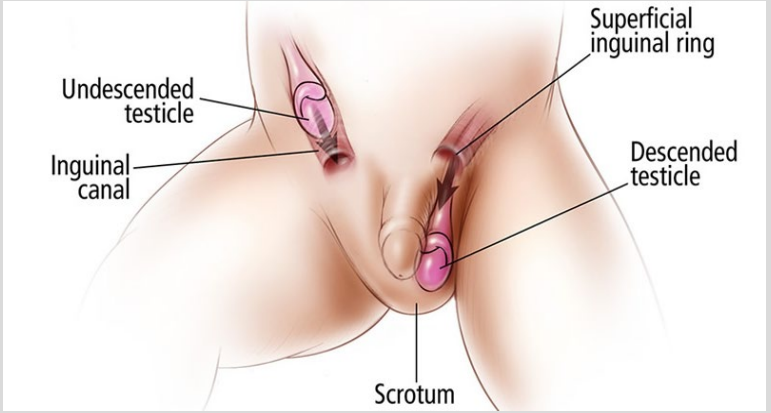


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Cryptorchidism: Clinical Symptoms

- Clinical symptoms of cryptorchidism include the following:
  - **Empty, underdeveloped, flat, rugated scrotum**
  - **Inguinal fullness** (may be initially mistaken for a hernia)
- Whenever you suspect cryptorchidism, you always want to look for clues suggesting endocrine or genetic causes due to the high association



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Cryptorchidism: Treatment

- Treatment for undescended testicles should occur **AFTER four months of age**
  - If the testicle is high in the inguinal canal AND has a possibility of descending on its own, you can monitor the patient and **wait to treat until 9-12 months of age**
- If the testicle has still not descended by 12 months, refer to urology
  - Treatment is usually exploratory surgery
  - Note: Hormonal treatment is NOT recommended
- Despite correcting the position of the testicle, the patient still has an increased risk for malignancy, infertility, inguinal hernias, and testicular torsion therefore the parents should be warned of these risks

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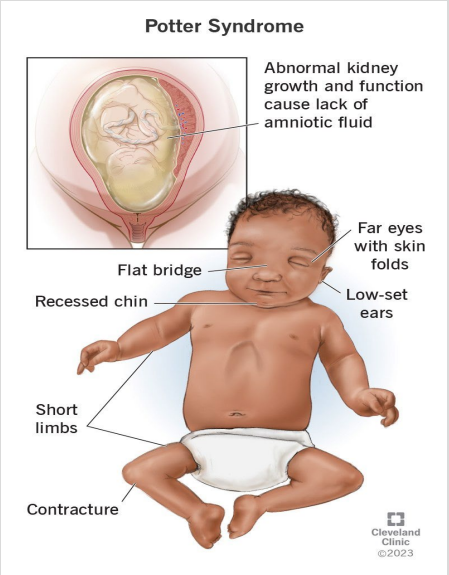
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Potter Syndrome

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Potter Syndrome: Overview

- Potter Syndrome occurs when **neonatal kidneys aren't functioning properly**—this is typically associated with renal agenesis, posterior urethral valves, or autosomal recessive polycystic kidney disease (PKD)
- Fetal urine is a large contributor to amniotic fluid
- Commonly presents with **oligohydramnios, pulmonary hypoplasia, and compression of the fetus** (e.g., club feet, abnormal facies, low set ears, and abnormal eye spacing)
    - Oligohydramnios occurs because the fetus won't produce enough urine which in turn creates a smaller volume of amniotic fluid—this can lead to compression of the fetus
- **Pulmonary hypoplasia causes significant mortality** because the lungs are too small and hypo-functioning



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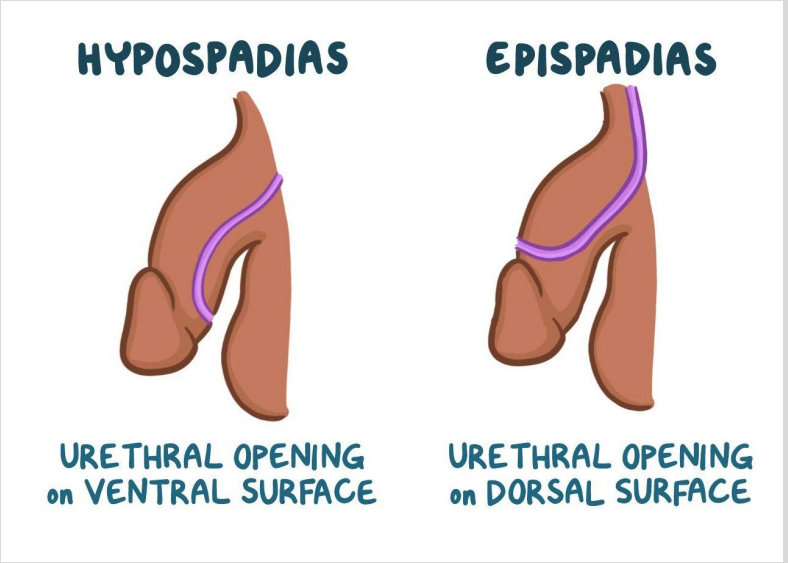
Other Pediatric Genitourinary Conditions

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Other Conditions: Hypospadias & Epispadias

- **Hypospadias** occurs when the opening of the urethra is anywhere along the **ventral shaft of the penis** rather than the tip of the glans
- **Epispadias** occurs when the opening of the urethra is anywhere along the **dorsal shaft of the penis** rather than the tip of the glans

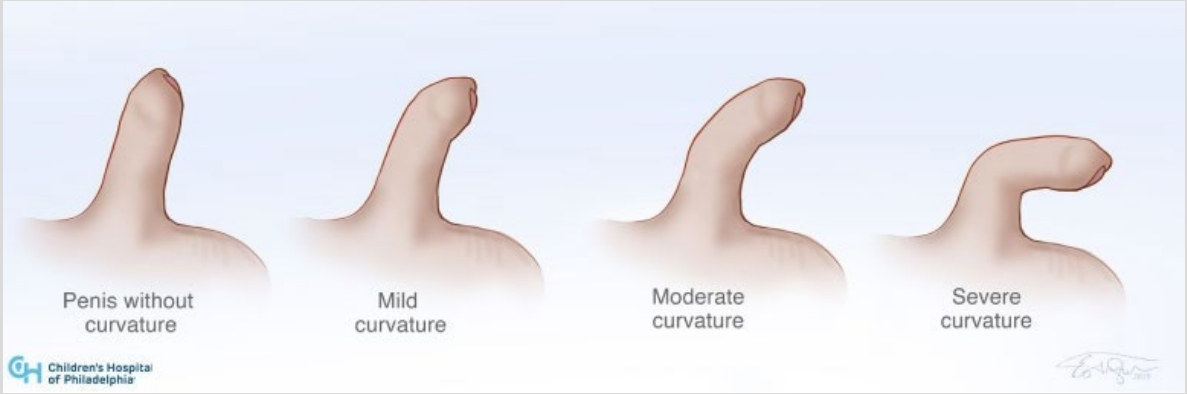
- It is important to NOT circumcise patients with hypospadias and epispadias as the foreskin is used to repair this condition



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Other Conditions: Chordee

- Chordee occurs when the penis curves downward because there is shortened skin on the ventral side (Note: It may also curve sideways)
- Can be corrected surgically
- This condition is often **associated with hypospadias and epispadias**

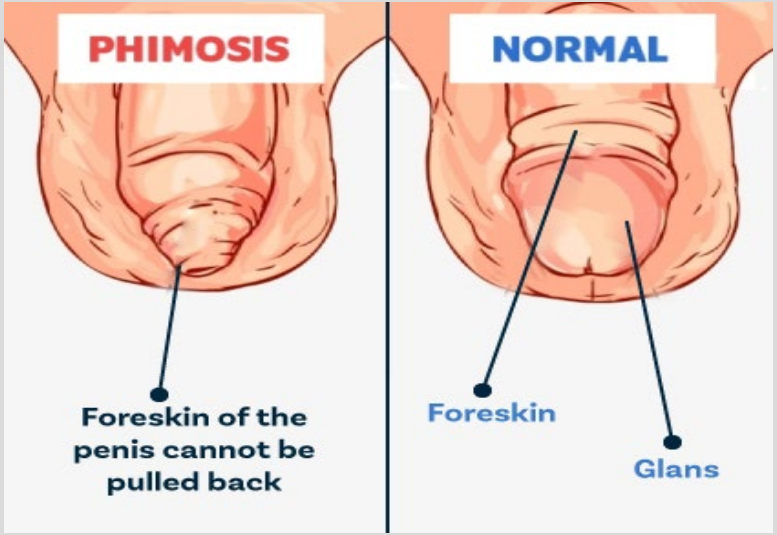


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Other Conditions: Phimosis

- Phimosis is the **inability to retract (ie. pull back) the foreskin of the penis**
- In newborns this may be **completely normal**—the foreskin will relax over time
- If the foreskin is intact and is unable to retract **DO NOT force it**
  - If forced to retract, the tightened foreskin may tear or get stuck which can cause vascular compromise



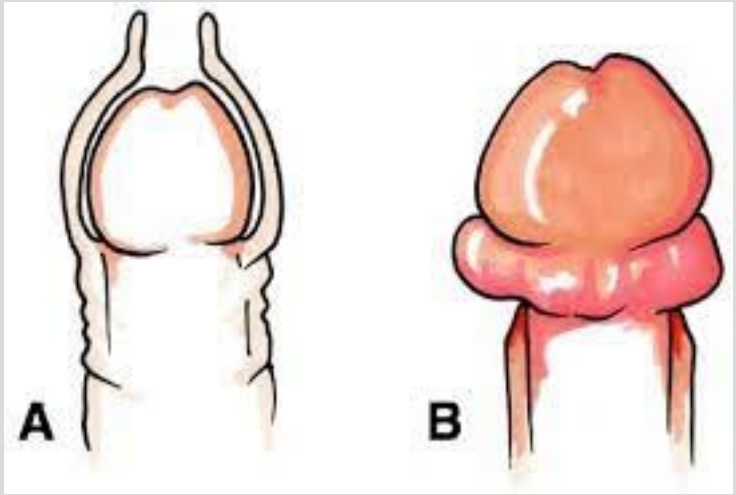
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Other Conditions: Paraphimosis

- Paraphimosis occurs when the **foreskin gets stuck behind the glans of the penis and cannot be put back into place** (ie. un-retracted)
  - The penis will look very red and inflamed (ie. big red angry portobello mushroom)
- This is a **medical emergency** as it is associated with **vascular compromise**—always refer to urology if you are unable to bring the foreskin back up

- If you are going to retract the foreskin you must always pull it back up!!!



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Other Conditions: Hair Tourniquet Syndrome

- Hair Tourniquet Syndrome is rather rare but occurs when a small piece of hair or string gets wrapped around an appendage
  - This appendage can be (most often) a finger or toe, but can also be a penis
  - This is a **medical emergency** as it can often cause vascular compromise
- The infant may often present as cranky/irritable, crying, and inconsolable



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Vaginal tag

- Normal
- Reassure parents

Pink, Mucous discharge also common in neonates (pseudo menses)



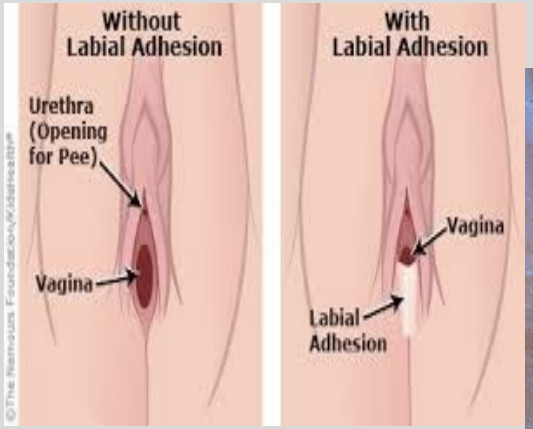
LABIAL ADHESIONS

- Relatively Common
- Prepubertal females
- Often will self resolve- esp in adolescence

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If mild and asymptomatic- no treatment

If symptomatic- Treat
dysuria, recurrent UTI's
vaginal pain/ discharge/ infections





- **Estrogen Cream**
- Steroid creams
- Physical separation (discouraged)

Condition	Key Clinical Pearls
Pyelonephritis	Diagnosis: Systemic Symptoms & Positive Urine Culture • Catheterized Specimen is Diagnostic Test of Choice Common Risk Factors: Vesicoureteral Reflux, Posterior Urethral Valve, Uteropelvic Junction Obstruction
Vesicoureteral Reflux	Failure of the bladder to empty properly resulting in buildup of urine in ureter and pelvis of kidney • Grades 1-2: Watch & Wait • Grades 3-5: Surgical Repair and Prophylactic Antibiotics
Posterior Urethral Valve	Piece of tissue partially blocking urethra resulting in weak stream • MALES ONLY • Results in BILATERAL hydronephrosis
Uteropelvic Junction Obstruction	Kink or anomaly in junction causes backflow of urine into ureter and kidney • UNILATERAL hydronephrosis
Cryptorchidism	At least 1 of the testicles does not descend properly through inguinal canal—presents as empty rugated scrotum • Wait to treat until 9-12 months of age • Associated with increased risk of malignant transformation, trauma (testicular torsion), inguinal hernia, and sub/infertility
Potter Syndrome	Neonatal kidneys aren't functioning properly resulting in oligohydramnios, pulmonary hypoplasia, compression of fetus (ie. club feet, abnormal facies, low set ears, abnormal eye spacing)
Hair Tourniquet Syndrome	Careful exam in inconsolable infant- Medical emergency → vascular compromise
Hypospadias & Epispadias	Hypospadias: opening of urethra is on the ventral shaft of the penis rather than the glans Epispadias: opening of urethra is on the dorsal shaft of the penis rather than the glans
Phimosis & Paraphimosis	Phimosis: inability to RETRACT foreskin of the penis Paraphimosis: inability to pull the foreskin of the penis back to normal position (this is a medical emergency!)

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Question #1:

A 35-year-old G2P2002 woman presents for a routine prenatal ultrasound at 20 weeks gestation. The ultrasound reveals multiple abnormalities, including bilateral renal agenesis and flattened facies in the developing fetus. There is also evidence of oligohydramnios. Which of the following conditions is most likely responsible for these findings?

A. Cystic Fibrosis
B. Congenital Adrenal Hyperplasia
C. Potter Syndrome
D. Trisomy 21
E. Fetal Alcohol Syndrome

Bonus: Name the complication of this condition that is associated with a high level of mortality.

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Question #2:

A 2-week-old male infant is brought into the emergency department with a high fever, irritability, and poor feeding. The parents report that the baby has been passing urine less frequently than usual with a weaker stream. Physical examination reveals a febrile and fussy infant. A renal ultrasound demonstrates bilateral hydronephrosis with dilated ureters. Which of the following congenital anomalies is the most likely cause of this infant's condition?

A. Vesicoureteral reflux
B. Posterior urethral valves
C. Uteropelvic junction obstruction
D. Paraphimosis
E. Hypospadias

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Question #3:

A 6-month-old male infant is brought to the pediatrician by his parents due to concerns about his genital development. The parents report that the infant's left testicle appears to be missing from the scrotum, and they have not observed any changes since birth. The infant is otherwise healthy and meeting developmental milestones appropriately. On physical examination, the right testicle is palpable in the scrotum, but the left scrotal sac is empty. The infant's genitals appear normal otherwise. Despite surgical repair, this infant will be at increased risk of what condition in the future?

A. Testicular Torsion
B. Subfertility
C. Testicular Malignancy
D. Inguinal Hernia
E. All of the above

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Question #4:

A 6-month-old male infant is brought to the emergency department by his parents due to a concerning genital condition. They report that while cleaning the baby's diaper area, they accidentally retracted the infant's foreskin behind the glans penis, and now they are unable to return it to its original position. The infant is visibly distressed and has developed swelling and discoloration of the glans. On examination, there is evident constriction of the foreskin behind the glans, preventing its reduction. Which of the following is the most appropriate initial management for this condition?

A. Immediate Circumcision
B. Manual Reduction of the Foreskin
C. Administration of Intravenous antibiotics
D. Application of topical corticosteroids
E. Observation and reassurance

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Lecture and Lab Feedback Form:

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